1- DIRECT TITRATION METHODS

In these methods the sample to be analyzed is titrated with a standard acid or base, usually prepared form strong electrolytes, according to the . nature of the sample. A suitable indicator has to be chosen according to the nature of the reactants. Direct titration methods are useful for the following:

A-Determination of Acids

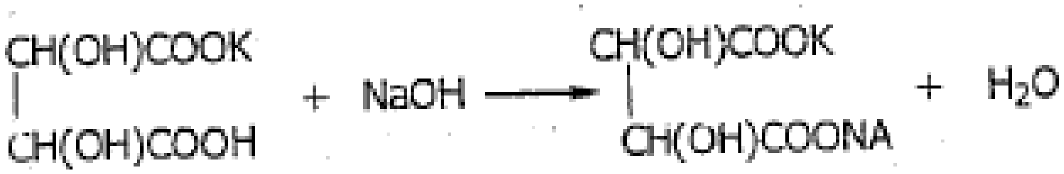
If standard alkali is available, we can determine the contents of various acids in solutions.

If the acid is **strong,** it is titrated in presence of methyl orange or phenolphthalein. In determination of **weak acids,** it is necessary to use phenolphthalein and not methyl orange. The acid should have Ka not less than 10^{-7} (pK_a < 7), if weaker, it cannot be titrated directly.

Acids possessing **limited solubilities** in water, e.g. benzoic, salicylic acids, etc should first be dissolved in neutralized ethanol. Water is then added and the acid now precipitated in finely divided condition is titrated with sodium hydroxide to phenolphthalein end point.

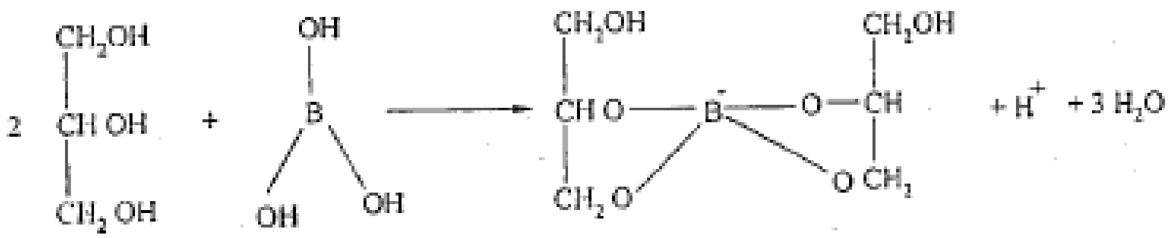
Free fatty acids, as in the determination of **"Acid Value"** of fixed oils, are first dissolved in ether-ethanol mixture, 1:1, previously neutralized to phenolphthalein, in which the oil itself is usually insoluble. The mixture is then titrated with potassium hydroxide until the first pink color of the indicator.

Acid salts, e.g. potassium acid sulfate KHS04 and potassium acid tartarate, can be titrated with standard sodium hydroxide using phenolphthalein.



As the potassium acid tartarate is sparingly soluble in water, titration should be carried out on hot solution.

Boric acid acts as a weak monobasic acid, Ka = 5.8 x 10^{-10}



The released protons, equivalent to boric acid, can be titrated against sodium hydroxide using phenolphthalein as indicator.

B Determination of Bases :

Titration of **strong bases** with standard acids presents no difference from the cases of inorganic acids.

For **weak bases**, titration with standard mineral acid solutions give salts, which will be hydrolyzed in aqueous solutions, and therefore, an indicator with a pH-range in the acid side is used (e.g. Methyl Orange). The end point in the titration of aminophylline solution with sulfuric acid is best indicated by bromocresol green. Pyridine solution can be titrated with hydrochloric acid to the full yellow color of bromophenol blue.

2- DOUBLE INDICATOR TITRATIONS

These are a type of direct titration but with the use of two indicators. They are used.for the determination of diprotic acids or mixtures of two monobasic acids.

I is possible to titrate mixture of two acids in two steps if the two conditions exist:

a) If the dissociation constant K_a for both acids is at least **10^{-7} (pK_a < 7)**

b) if the ratio of their dissociation constants Ka_1/Ka_2 is high (at least 10^4) OR the difference in their pK_a is at least 4.

This condition is essential for a sharp break on titration curves.

HCl/ acetic acid mixture:

Therefore, it is possible to titrate HCI in presence of acetic acid, Ka=S.Sx10-s. The H+ of HCI suppresses the ionization of the other weak acid, by common ion effect, so that sodium hydroxide neutralizes HCI first.

$$HCl \rightarrow \boxed{H^+} + Cl^-$$
$$CH_3COOH \rightleftharpoons \boxed{H^+} + CH_3COO^-$$

Methyl orange is used in this step as indicator. The weak acid is tt1en determined by titration with the alkali using phenolphthalein as indicator.

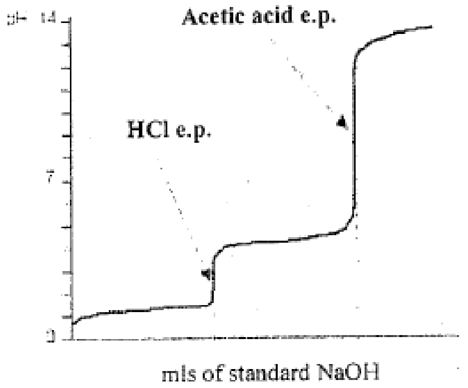


Fig. 6: Titration curve of HCl/CH3COOH mixture by standard NaOH

With **diprotic acids,** the molecule contains two replaceable hydrogen atoms. The ionization occurs in successive steps, each reaction having a characteristic ionization. Thus, for a dibasic acid H_2A:

$H_2A \rightleftharpoons H^+ + HA^-$, $K_1 = [H^+][HA^-]/[H_2A]$

$HA^- \rightleftharpoons H^+ + A^{2-}$, $K_2 = [H^+][A^{2-}]/[HA]$

The ratio of Ki to K2 must be at least 10^4 • f less, the inflection at the titration curves are smeared out and only one break is observed, corresponding to neutralization of both hydrogen ions.

For separable dissociation constants of a diprotic acid, the pH and consequently the appropriate indicator of the first neutralization step correspond to:

$$\mathbf{PH_1 = 1/2\ (PK_1+PK_2)}$$

While the pH corresponding to the second step is calculated from the equation of salt of weak acid-strong base:

$pH_2 = 1/2 pKw + 1/2 pK_2 - 1/2 pCs$

And an indicator is selected accordingly.

H_2SO_4

It has two dissociation constants, they are near each other that all available H+ can only be titrated in one step (Ka2/Ka1 is less than 10^{-4}).

K_1 = strong acid, $K_2 = 1.15x10^{-2}$

<u>Carbonic acid</u>

$H_2CO_3 \rightleftarrows HCO_3^- + H^+$ $K_1 = 4.3x10^{-7}$, $pK_1 = 6.37$

$HCO_3^- \rightleftarrows col_3^{2-} + H^+$ $K_2 = 5.6x10^{-11}$, $pK_2 = 10.25$

Its titration curve with sodium hydroxide has two equivalence points:

$H_2CO_3 + NAOH \longrightarrow NaHCO_3 + H_2O$

$NaHCO_3 + NAOH \longrightarrow Na_2CO_3 + H_2O$

We can calculate the pH of the two equivalence points:

$PH_1 = \frac{PK1+Pk2}{2} = 8.13$

$PH_2 = 1/2\ PK_w + 1/2\ PK_a - 1/2\ PC_s = 11.6$

$PK_2 > 7$, So the second step cannot be titrated, and the ratio K_1 / k_2 is less than 10^4 (0.8×10^4). Therefore, direct titration of carbonic acid as a dibasic acid is clearly impossible. Carbonic acid, therefore, is titrated as monmbaic acid in presence of phenolphthalein (PH_1 is within the range of phenolphthalein).

Hydrogen sulfide

$H2S \rightleftarrows HS^- + H^+$ $\quad K_1 = 9.1 \times 10^{-8}$ ($PK_{a1} = 7.04$)

$HS^- \rightleftarrows S^{2-} + H^+$ $\quad K_2 = 1.2 \times 10^{-15}$ ($PK_{a2} = 14.92$)

There is a sufficient difference between the two Ka (K1/k2 is more than 10^4) but both hydrogens are so weak to be directly titrated (K1 and k1 are less than 10^{-7}).

H2S might, therefore, be determined indirectly by precipitating the sulfide ion with a cation such as copper, which would make the two protons available for reaction with alkali.

$$H2S + Cu^{2+} \rightarrow CuSc(s)\downarrow + 2H^+$$

<u>Phosphoric add</u>

The same above rules will apply to a polyprotic acid, H_3A. The following reactions take place if phosphoric acid is titrated with sodium hydroxide.

$$H_3PO4 + NaOH \rightarrow NaH_2PO_4 + H_2O$$

$$NaH_2PO_4 + NaOH \rightarrow Na_2HPO_4 + H_2O$$

$$Na_2HPO_4 + NaOH \rightarrow Na_3PO_4 + H_2O$$

Accordingly, phosphoric acid titration curve has not one but three equivalence points. The numerical value of the dissociation constants of phosphoric acid is:

H3PQ4 H2PQ4- + H+ (pKa1 = 2.12)

H3PQ4- HP04^{2-} + H+ (pKa2 = 7.21)

HP0^{2}-4 4 =-... PQ4^{3-} + H+ (pKa3=12.67)

The pH of solutions of acid salts, as shown above, is the arithmetic mean of the two dissociation exponents of the corresponding acid. The position of the first equivalence point, therefore, is found from the equation:

pH1 = (pK1+pK2)/2 = (2.12+7.21)/2 = 4.66 "M.0"

The second equivalence point occurs at pH:

pH2 = (pK2+pK3)/2 = (7. 12+ 12.67) *I* 2 = 9.94 "Ph.Ph."

pH3 can be calculated from the formula for pH of salts of weak acids strong bases:

pH3=7+ ½pKa- ½pCs

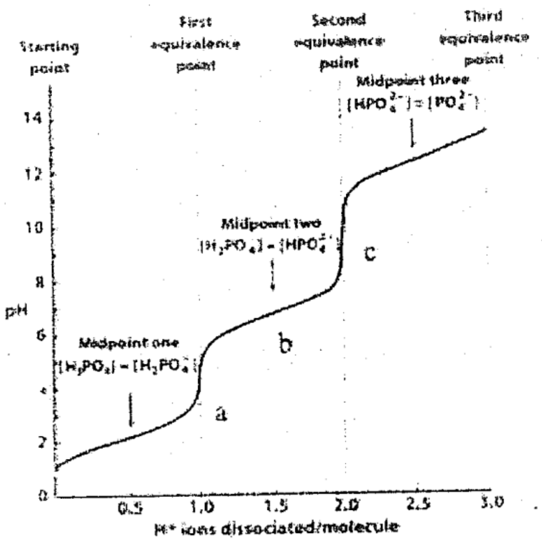


Fig. 7: Titration curve of 50ml 0.1N phosphoric acid with 0.1N NaOH.

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It is clear from calculations and from figure 7 that pH breaks are large enough to allow accurate titration of phosphoric acid at the first and second equivalence points using methyl orange and phenolphthalein as indicators, respectively.

Direct titration of orthophosphoric acid, as a tribasic acid is impossib!e with aoy1ndicato.r as K3 is.very small,. 2.2x10-^{13}. Such titration, therefore, may be done indirectly.

Phosphoric acid is replaced by an equivalent amount of HCI by adding neutral CaCb..The liberated equivalent amount of HCI is then easily titrated.

2 HJP04 + 3CaC'2 $\longrightarrow$ Ca3(P04)2t + 6HCI or

2 Na2HPQ4 + CaC'2 $\longrightarrow$Ca3(PQ4)2t + 4NaCI + 2HCI

Citric add

On the other hand, citric acid has the following three stages of dissociation constants:

$Ka1=9.2x10^{-4}$ pKa1 = 3.04

Ka2= 2.7xlo-s pka2 = 4.57

Ka3=1.3x1 Q^{-6} pKa3 = 5.89 .

These are close together that the three stages cannot be differentiated. As K3 is about 10^{-6}, then all the hydrogen can be determined in one step using a suitable indicator.

3- DISPLACEM ENT TITRATIONS

Easily hydrolysable salts formed from:

1. Salts from a strong base and a weak acid, e.g. potassium cyanide, borax and sodium carbonate.
2. Salts from a strong acid and a weak base, e.g., ferric chloride and aluminium sulfate.

Potassium cyanide

$$KCN + H_2O \rightleftharpoons HCN + K^{+} + OH$$

On adding hydrochloric acid, the reaction takes place with the OH- produced by hydrolysis, the equation of such a reaction will be:

$$KCN + HCI \longrightarrow HCN + KCI$$

The net result is that weak HCN has been displaced by the stronger HCI. The process is, therefore, often referred to as a "displacement titration".

At the start of titration, the pH is calculated using the formula of salt,

$$\mathbf{pH = 7 + 1/2pKa - 1/2\,pCs}$$

During the titration the solution contains the free weak acid together with residual unconverted salt and pH is calculated using the formula for acidic buffers:

pH = pKa + log [Salt]/[Acid]

At the equivalence point, all the salt is converted into the free acid, its pH can be calculated using formula of weak acid:

pH = 1h pKa + V2Ca

Finally, when the titrating acid is present in excess the solution pH is calculated in the usual way, from the total HCI concentration in solution.

The equivalence point of the titration lies in the acid region. Accordingly, such salts can be titrated in presence of methyl orange or methyl red indicators, but not in presence of phenolphthalein.

<u>Sodium carbonate and bicarbonate</u>

Na2C03 + 2H20 $\longrightarrow$ 2NaOH + H2CQ3

Na2HC03 + H20 $\longrightarrow$ NaOH + H2C03

Following the same principle, the strong acid displaces the week one in its salt, they can therefore be estimated by displacement titration with a mineral acid as HCL

Na2C03 + HCI $\longrightarrow$ NaHC03 + NaCl

The pH of the first equivalence point is calculated to be 8.3, i.e. within the range of phenolphthalein indicator. This solution can further be titrated with hydrochloric acid until the second equivalence point.

NaHCQ3 + HCI $\longrightarrow$ H2C03 + NaCl

At this stage neutral sodium chloride will be formed together with the very weak carbonic acid. The pH of the solution drops to about 3.8 and methyl orange can be used as indicator for the second equivalence point.

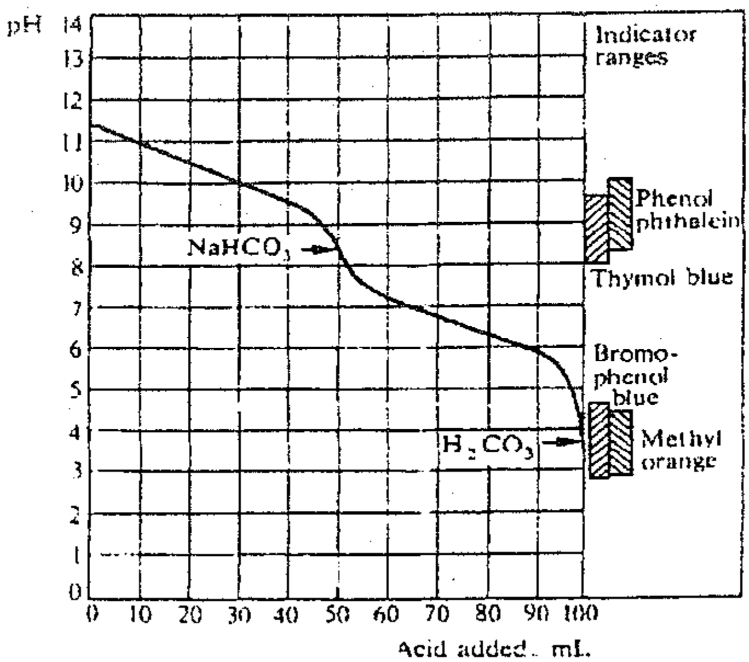


Fig. 8: Titration curve of Sodium Carbonate Solution with Hydrochloric Acid.

	$CO_3^{--} \xrightarrow{H^+}$	$HCO_3^- \xrightarrow{H^+}$	$CO_2 + H_2O$
pH	11.7	8.3 ½ neutralization	3.8
Ph.Ph.	Alk.	Acidic	Acidic
M.O	Alk.	Alk.	Acidic

The first half – neutralization step, however, is supposed to take Place : in two separate steps:

$Na2C03+2HCI \longrightarrow 2NaCl+CO_2+H_2O$ ……….(1)

$Na2C03+CO_2+H_2O \longrightarrow 2NaHCO3$ ….…….(2)

(1) + (2) $2Na2C03+2HCl \longrightarrow 2NaCl+2NaHCO3$

n the titration of sodium carbonate to half-neutralization step, therefore certain **precautions** have to be taken into consideration to avoid the escape of C02:

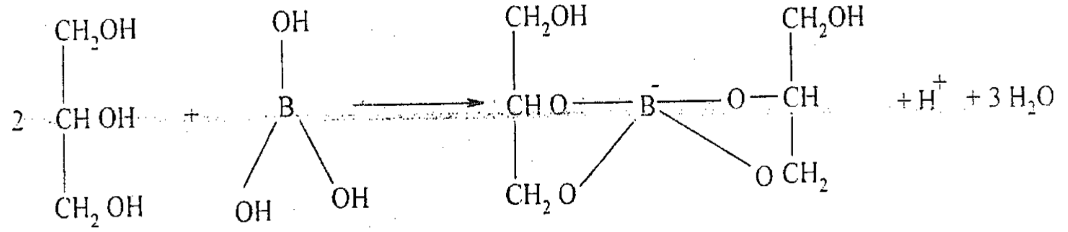
- Big dilution of the titrated solution.
- Dipping the burette nozzle below the surface of the solution.
- Cooling.
- Slow titration
- Continuous stirring.

Borax

$Na_2B_4O_7+7H_2O \rightleftarrows 4H_3BO_3+2NaOH$

As boric acid is a very weak acid, the produced sodium hydroxide can be directly titrated with standard hydrochloric acid to methyl orange end point.

The neutralized solution can then be used for the determination of boric acid after the addition of an equal amount of glycerol:



Glyceroboric acid can then be titrated with sodium hydroxide to phenolphthalein end point.

4H3BQ3 + 4NaOH $\longrightarrow$ 4NaB02+8H20

When a pure sample of borax, therefore, is titrated the volume of standard 21lkali used would be exactly double that of the standard acid, of the same normality.

If a borax solution is treated with glycerol, the solution can be regarded as a half-neutralized boric acid solution

Na2B4Q7 + 3H20 +glycerol $\longrightarrow$ 2NaB02+2H3B03

Borax may sometimes be contaminated with boric acid or sodium carbonate.

Borax/Boric acid mixture

1- The acid reading using methyl orange as indicator will correspond to the borax and can be used directly for its calculation.

2- The addition of glycerol to the previous neutralized solution

will potentiate the boric acid allowing its titration with a standard alkali using phenolphthalein. This boric acid is coming from borax hydrolysis in water and the originally present free boric acid.

Borax = rnl1

Free boric acid = ml2 - (2 x ml1)

<u>Na2C03 / Borax mixture</u>

1- Titration with standard acid to methyl orange end point is a measure of total alkalinity, i.e., all $C03^{2-}$ and oH^- "from borax".

2- The previous solution is boiled to remove carbon dioxide (phenolphthalein is sensitive to it).

An equal volume of glycerol is then added and the glyceroboric acid is titrated with standard alkali using phenolphthalein.

Borax = ml2

CO_3^{2-} = ml1 – ½ ml2

Carbonate/Bicarbonate mixture

1- The total is determined by titration with standard acid, using methyl orange as indicator.

2- The $C03^{2-}$ is determined by titration with standard acid, using phenolphthalein as indicator and taking the above-mentioned precautions to guard against escape of C02.

C03^{2-} = 2 ml2

HC03- = ml1 - 2 ml2

Na2C03/NaOH mixture

1- Total is determined by acid titration using methyl orange as indicator.

2- Acid titration using phnoJphthaleln as indicator is a measure of all OH^- and 1/2 co_3^{2-}

$C0_3^{2-} = 2 \times (ml_1 - ml_2)$

$OH^- = 2\ ml_2 - ml_1$

OH^- *I* HCO_3^- mixture

They can't exist together

$OH^- + HCO3^- \rightarrow H_2O + CO_3^{2-}$

Hydrolysable add salts

The same principle, as in basic salts, is used in estimating the free acid in hydrolysable acid salts such as ferric chloride, aluminum chloride or sulfate, ect.

$AICl_3 + 3H_2O \rightleftarrows Al(OH)_3 \downarrow + 3HCI$

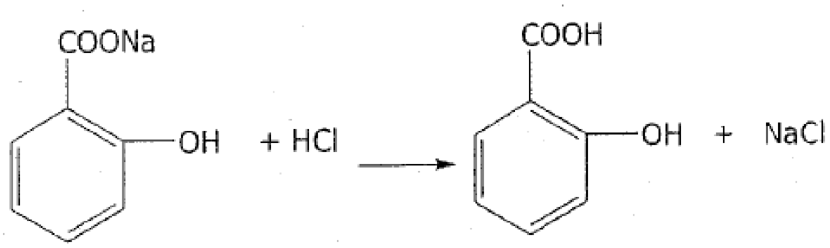
A precipitate of the insoluble metal hydroxide will be produced. A complex-forming anion therefore has to be added to from a stable complex with the metal, leaving the free acid to be titrated.

Aluminum chloride is complexed with neutral produced oxalate and the liberated hydrochloric acid is titrated with standard sodium hydroxide using phenolphthalein as indicator.

4- BIPHASIC TITRATIONS

The method is applied for water soluble salts, the acid of which is insoluble in water but soluble in other immiscible solvent Example of such salts determined by this method is alkali metal and ammonium salicylate and alkali metal benzoate.

Sodium salicylate



The liberated salicylic acid is so strong to give an acidic pH and this affect the indicator. It must, therefore1 be removed as soon as it is produced.

The idea depends on the greater solubility of salicylic acid in ether, 250 times greater than in water. To further minimize this solubility, one use 3 volumes of ether as the aqueous layer. Such a titration is done with standard acid to bromophenol blue end point, using separating funnel (colour change from bluish green to pale green).